

Email Campaign Experiment

Decision memo for a historical 64,000-customer, three-arm randomized email campaign.

Decision

Treat **Mens E-Mail** as a candidate for fresh validation based on its observed lift in this historical experiment. A new rollout requires evidence on incremental contribution, delivery cost, unsubscribe risk, and deliverability.

Evidence

ITT contrast	Visit lift	Conversion lift	Spend lift (currency/customer)
Mens E-Mail vs No E-Mail	+7.66 pp [7.00, 8.32]	+0.68 pp [0.50, 0.86]	+0.77 [0.49, 1.05]
Womens E-Mail vs No E-Mail	+4.52 pp [3.89, 5.16]	+0.31 pp [0.15, 0.47]	+0.42 [0.17, 0.68]
Mens vs Womens E-Mail	+3.14 pp [2.43, 3.84]	+0.37 pp [0.17, 0.56]	+0.35 [0.03, 0.66]

Brackets are pointwise, unadjusted large-sample 95% confidence intervals, not simultaneous bands. pp = percentage points; spend is in dataset currency units per assigned customer. The six email-versus-control p-values were Holm-adjusted together; the three Mens-versus-Womens p-values were adjusted as a separate family. All were below 0.05 after adjustment. Maximum email-versus-control baseline |SMD| was 0.016.

Interpretation

Both email treatments increased visits, conversions, and mean spend relative to control in this experiment. Mens E-Mail had the larger observed effect on all three outcomes. One subgroup check comprised two visit-effect interactions by newbie status, with Holm correction; neither was significant. This does not establish equal effects across subgroups.

Launch condition and limitations

The data contain revenue, not profit, and omit campaign cost, gross margin, unsubscribe risk, and deliverability. Under a simplified model with fixed gross-margin rate $m \geq 0$ and cost c per assigned customer, the Mens-versus-control pointwise lower revenue endpoint gives a contribution sensitivity bound of $m \times 0.485 - c$. It is positive when $c < m \times 0.485$; this is not a selection-adjusted or simultaneous profit guarantee. A new rollout should retain a randomized holdout and define contribution and customer-experience guardrails in advance.

Method: intention-to-treat mean differences with unpooled standard errors. Without customer IDs, repeated rows cannot be classified as repeated customers and are retained. Historical results do not establish effects for a new population or channel. Public data: [MineThatData E-Mail Analytics Challenge](#). Raw data are not redistributed.